Custom Suction Cup Gripper
Hardware Mechanical Engineering Project
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Introduction

The task is to design a suction cup gripper to grab foil bags. This written composition will go through the component selection, discuss my design choices, and provide references and supplementary materials. The goal of the project was to create a professional robotic gripping solution that satisfies the customer's requirements while maintaining compatibility between all components.

The customer requires a robot capable of working alongside humans with a minimum reach of 1200 mm and a minimum payload of 10 kg. The robot must also utilize the customer's existing 6 bar compressed air connection to power the suction gripping system. Based on these requirements, a collaborative robotic system with a vacuum-based end-of-arm tool was designed.

The final system consists of a collaborative robotic arm, an automatic tool changer, a vacuum generator, a suction cup designed for foil bag handling, and a level compensator to improve gripping reliability. A custom gripper bracket was designed in CAD (Fusion 360) to integrate these components into a complete and compatible assembly.

Component Selection

I. Robot

The robot chosen for this task is a UR15 collaborative robot from Universal Robots. This robot is suitable to work alongside humans because collaborative robots are designed with built-in safety systems such as force and torque sensing that allow them to detect unexpected contact and stop motion when necessary.

The UR15 has a reach of 1300 mm, which exceeds the requirement of 1200 mm. It also has a payload of 17.5 kg, which exceeds the required payload of 10 kg. This ensures that even after accounting for the weight of the gripper assembly and tooling components, the robot can safely lift the foil bags while remaining within its operational limits. The estimated mass of the end-of-arm tooling, including the tool changer, vacuum generator, custom bracket, and suction components is approximately 3-4 kg. This still leaves a comfortable payload margin within the UR15's 17.5 kg capacity.

Another advantage of the UR15 is its standardized ISO tool flange, which simplifies integration with robotic tooling such as automatic tool changers and vacuum grippers.

II. Gripper Changing System for the Robot

An SMC RMTA Automatic tool changer was chosen for this project. The main plate selected has a 20 kg payload capacity without a connector, and the tool plate also has a 20 kg payload capacity with a female threaded mounting interface and no connector. This payload

capacity was chosen to ensure the tool changer can safely support the gripper and the bag payload with an adequate safety margin.

An automatic tool changer was selected because it allows the robot to automatically connect and disconnect different end-of-arm tools when necessary. This improves flexibility and efficiency in an industrial environment where multiple tools may be used.

The RMTA system also includes integrated pneumatic pass-through channels, allowing compressed air to pass through the tool changer from the robot side to the tool side. This is important for this application because the customer's 6 bar compressed air supply is used to power the vacuum generator that drives the suction cup gripper.

III. Vacuum Pump for Bags

The vacuum pump used for this system is a Piab COAX MIDI Si32-3 vacuum generator. This device generates vacuum from compressed air using the Venturi principle.

Compressed air enters the generator through the inlet port. As the air passes through the internal nozzle, it accelerates and creates a pressure drop that generates vacuum at the suction port. This vacuum is then transmitted through the suction line to the suction cup, allowing it to grip the foil bag.

The COAX vacuum generator was selected because it is designed for applications where there may be surface leakage, such as flexible packaging or foil bags. Foil bags often have wrinkles and uneven surfaces that make it difficult to maintain a perfect seal. The multi-stage COAX design provides high initial vacuum flow, which helps quickly establish suction even when minor leakage occurs.

IV. Suction Cup to Lift Bags

The suction cup selected for this design is a Piab piGRIP BGI80 suction cup. This suction cup is specifically designed for handling bags and flexible packaging materials.

The BGI bag lip design allows the suction cup to adapt to the shape of flexible foil bags. Unlike flat suction cups, which require a perfectly smooth surface, the bag lip design can maintain suction even when the surface of the bag is wrinkled or uneven.

The suction cup has three bellows, which allow the cup to flex and adjust when contacting the bag. This flexibility improves sealing performance and reduces the likelihood of losing grip during handling.

The suction cup also has a diameter of 80 mm, which provides sufficient surface area to generate the vacuum force required to lift the bags. Because the suction cup only needs to seal a portion of the bag rather than the entire surface, it is a reliable choice for this application.

Additional details about the suction cup configuration are provided in the supplementary information section.

V. Level Compensator

A Piab LC25-M3840 level compensator is used in this gripper assembly. The level compensator provides 40 mm of vertical stroke, allowing the suction cup to move up and down relative to the gripper body. Choosing the male connection for this level compensator makes it

compatible with the female threading on the suction cup, eliminating the need for additional adapters and reducing potential vacuum leakage points.

This movement allows the suction cup to compensate for variations in bag height and surface irregularities. When the robot lowers the suction cup onto the bag, the compensator compresses slightly, allowing the suction cup to settle onto the surface and form a reliable vacuum seal.

The level compensator also helps absorb impact forces during contact, protecting the robot wrist, the suction cup, and the foil bag from sudden loads.

Engineering Validation of Suction Cup Selection

To further support the suction cup selection, I performed a simplified vacuum holding force check. The purpose of this calculation is not to replace physical testing, but to confirm that the selected suction cup size is reasonable for the application.

Vacuum holding force can be estimated using the relationship:

$$F = P \times A$$

where F is the holding force, P is the vacuum pressure difference, and A is the effective suction area. Assuming a conservative vacuum level of 60 kPa and an 80 mm diameter suction cup, the suction area is approximately

$$A = \pi \times (0.04m)^2 = 0.00503\ m^2$$

This gives a theoretical holding force of

$$F = 60{,}000\ Pa \times 0.00503\ m^2 = 301.8\ N$$

The weight of a 10 kg bag corresponds to approximately 98 N of gravitational force. The calculated holding force of 301 N therefore provides a theoretical holding margin of roughly 3:1 relative to the 98 N load before safety factors are applied. Since flexible foil bags can have wrinkles, leakage, and shifting loads, a safety factor must be applied. Using a conservative safety factor of 3 to 4, the effective holding capacity is reduced significantly, which confirms why a bag-specific suction cup with a larger diameter was the most appropriate choice for this single-cup design. This calculation supports the decision to use the Piab piGRIP BGI80 rather than a smaller suction cup.

Explanation of CAD

The CAD assembly was created to integrate all selected components into a functional end-of-arm tool.

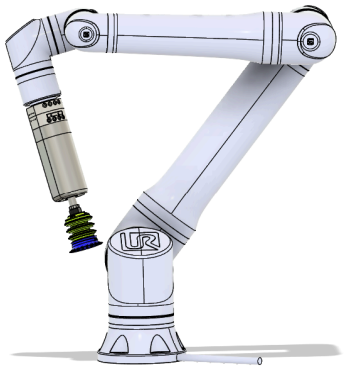

Figure 1

Figure 2

Figure 3

The assembly consists of the UR15 robot arm, the SMC RMTA tool changer, the custom-designed gripper bracket, the Piab COAX vacuum generator, the LC25 level compensator, and the BGI80 suction cup forming the end-of-arm tool. Figure 1 shows the full robot assembly, Figure 2 shows the custom gripper with all the components inside, and Figure 3 shows the internal components, including the vacuum pump and the level compensator.

The custom gripper bracket serves as the main structural interface between the tool changer and the vacuum gripping components. The bracket mounts directly to the RMTA tool plate and provides mounting features for both the level compensator and the vacuum generator.

The level compensator and suction cup are mounted on the centerline of the gripper to ensure that the load from the foil bags is aligned with the robot's tool center point and approximately centered with the tooling center of gravity to minimize wrist torque during lifting. This improves stability during lifting operations. Care was also taken to ensure that the level compensator's full 40 mm stroke can occur without interference with the gripper bracket or adjacent components.

The vacuum generator is mounted on a side extension of the bracket. This placement prevents interference with the vertical movement of the level compensator. Locating the vacuum generator directly on the gripper bracket also reduces vacuum line length, which reduces vacuum response time and helps achieve faster pick cycles.

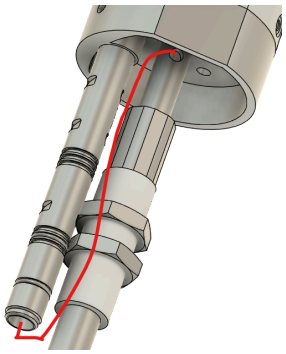

Figure 4

Figure 5

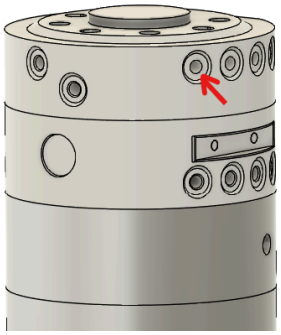

Figure 6

Two pneumatic lines are used in the design. One carries compressed air from the tool changer to the vacuum generator, as shown in figure 5. The second line carries vacuum from the generator to the suction cup assembly, as shown in figure 4. The customer's 6 bar compressed air connects to the main plate of the tool changer as shown in figure 6. This layout ensures efficient vacuum generation and reliable gripping performance.

Overall, the final design satisfies the functional requirements while incorporating practical engineering considerations such as payload margin, pneumatic routing, sealing reliability, and manufacturable end-of-arm tooling design.

Supplementary Information

The total time it took to complete this project was approximately 16 hours. This includes component research, system design, CAD modeling, and documentation.

With additional time, further improvements could include optimizing the gripper bracket design, integrating a more detailed pneumatic manifold, and performing additional calculations to validate the vacuum holding force required to lift the foil bags. Additionally, component and assembly testing will help improve this project. I designed the custom gripper to include 2 parts for easier assembly, and future iterations of this design will include mounting holes to attach the parts together and will ensure they are manufacturable. Future iterations would also evaluate structural stiffness of the bracket and potential vibration effects during rapid robot motion. Throughout this design, the CAD I modeled contains allowances for parts to be slip fitted.

Characteristics of the suction cup used:

- Certification: None
- Lip type: Bag lip with retainer
- Lip size: Ø 80mm
- Lip material and hardness: Silicone 50° Shore A, Blue
- Bellows: 3 folds
- Support: Support S1
- Fitting type: G-Thread
- Fitting size: 3/8"
- Fitting style: Female
- Quick selection: No Option
- Filter: None
- piSAVE®: No piSAVE

Component reference: 0201636 - Ø 80 mm Bag lip, Silicone 50° Shore A

Other components considered:

Several alternative components were considered during the design process.

For the robot selection, other collaborative robotic arms were evaluated. A UR12e was considered due to its cost efficiency, however its payload capacity of 12.5 kg would leave limited margin once the weight of the gripper assembly is included, making the UR15 a safer choice.

For the tool changer, an SMC RMTA1-10 model was also considered. However, this model did not provide sufficient payload capacity for the intended application.

For the suction cup, a PSPF 53 SI-30/55 G3/8-IG suction cup was considered. Its smaller diameter would not provide enough surface area to reliably lift a 10 kg foil bag using a single suction cup. This option may have been viable if the design incorporated multiple suction cups instead of a single cup.

References

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Web catalog ｜SMC corporation. https://www.smcworld.com/webcatalog/en-jp/seriesList/?id=RMTA-E&show_page=true&type=GUIDE